

My Magic Map AGI

Embodied world exploration meets Radeon AI

A body-controlled Google Street View and 3D Maps experience that turns real places into artwork, postcards and virtual memories, and understands pointed-at objects with AMD Radeon GPU acceleration.

11.981 s

WARM SDXL AVERAGE

Three measured image-to-image runs

7.089 s

WARM QWEN VISION

Three uncached pointed-object runs

24.00 GiB

MEASURED PEAK VRAM

Qwen and ComfyUI resident together

48 GiB

RADEON WORKSPACE

gfx1100 - ROCm/HIP 7.2

Application: My Magic Map AGI

Participant and team: Eimis Pacheco

Submission repository: github.com/EimisPacheco/my-magic-map-agi-adm

Prepared August 2026

A map that responds to the whole person

Problem, target users and practical application scenarios

PROJECT BACKGROUND

Most digital maps are passive visual tools controlled by a mouse. My Magic Map AGI turns geographic exploration into an embodied, multimodal experience. A webcam becomes a private local sensor: walking in place advances through Street View, body direction steers, deliberate arm poses control flight, and hand, nose or companion-paw pointing selects visible objects.

The Track 1 adaptation adds AMD-first creative image generation and semantic vision. Users can transform the exact place they are viewing into artwork, postcards and virtual memories, or ask what a pointed-at object is.

ACCESSIBLE EXPLORERS

Hands-free navigation and large physical gestures make map discovery more approachable.

TRAVELERS AND FAMILIES

Create visual memories and postcards grounded in recognizable real destinations.

STUDENTS AND EDUCATORS

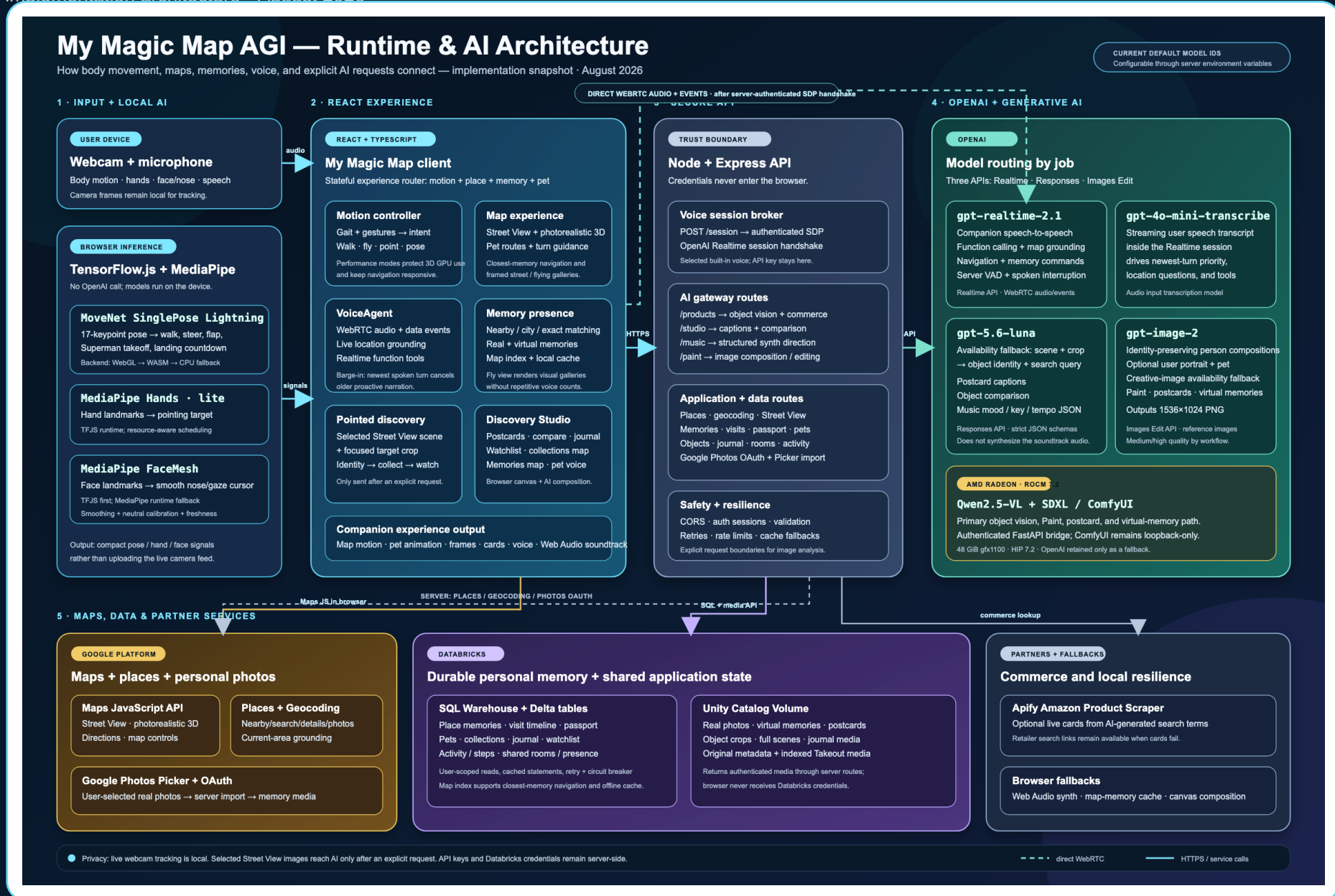
Combine geography, embodied learning and object discovery in one interactive environment.

CREATORS

Turn a chosen Street View scene into multiple distinct, downloadable artistic outputs.

Deliberate split: local motion, secure APIs, Radeon inference

Current implementation architecture - August 2026



Five sensing and AI layers with clear ownership

The GPU is used where throughput materially improves the experience

BROWSER MOTION

- MoveNet SinglePose Lightning: 17-keypoint pose for walking, steering and flight intent.
- MediaPipe Hands: hand landmarks for explicit target selection.
- MediaPipe FaceMesh: face orientation, expressions and nose pointing.
- GaitEstimator and FlyingDetector: deterministic temporal state machines.

RADEON CREATION

- SDXL base 1.0 through ComfyUI and PyTorch/ROCm.
- Image-to-image workflow preserves recognizable place structure.
- Prompt, palette, CFG, denoise, sampler and seed are controlled by the server workflow.
- Used by Paint a Place, standard postcards and virtual memories.

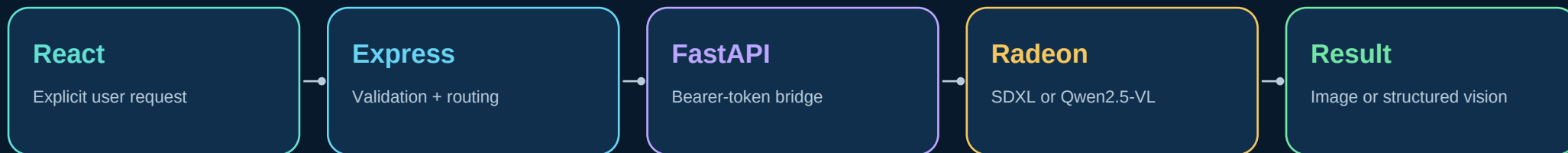
RADEON VISION

- Qwen2.5-VL-3B-Instruct in BF16 on the Radeon GPU.
- Receives the complete Street View scene plus a target-centered crop.
- Returns validated structured identity, category, confidence and safe search terms.
- Used for hand, nose and companion-paw object discovery.

OpenAI boundary: Realtime voice remains an OpenAI service. OpenAI is also an availability fallback and the current identity-preserving two-reference fallback. Gemini is not used.

From hosted generation to an AMD-first multimodal service

Secure, measurable and reproducible GPU execution



ADAPTATION DETAILS

- AMD OneClick base container with ROCm 7.2.1 and PyTorch 2.9.1+rocm7.2.1.
- Checksum-verified SDXL checkpoint stored in persistent model storage.
- ComfyUI stays on 127.0.0.1; only the authenticated bridge is tunneled.
- The repository versions the FastAPI bridge and exact ComfyUI workflow.
- The app records upload, GPU generation and total round-trip timing.

WHY THIS GPU BOUNDARY

- Live webcam frames remain local for privacy and immediate motion response.
- Selected Street View imagery reaches AI only after an explicit user action.
- Image generation and semantic vision are the highest-value GPU workloads.
- One Radeon service supports complementary creative and understanding tasks.
- Fallbacks preserve availability without overstating which workloads run on AMD.

Three complete user workflows

Each path starts with a real place and ends with an actionable artifact

PAINT A PLACE

- 1 Navigate with body motion
- 2 Select the visible Street View scene
- 3 Choose style and palette
- 4 SDXL image-to-image on Radeon
- 5 Display provider and timing
- 6 Download or save the artwork

POSTCARD / MEMORY

- 1 Choose a discovery or destination
- 2 Generate a distinct creative direction
- 3 Radeon SDXL produces the image
- 4 Add caption and place context
- 5 Persist to the memory system
- 6 Reopen from the gallery or map

POINTED VISION

- 1 Point with hand, nose or companion paw
- 2 Capture scene and focused crop
- 3 Qwen2.5-VL runs on Radeon
- 4 Validate structured identity
- 5 Show confidence and uncertainty
- 6 Collect, compare or search safely

Actual execution on gfx1100 with ROCm/HIP 7.2

Representative Street View scene - verified August 6, 2026

Measurement	Result	Evidence boundary
Radeon GPU	AMD Radeon Graphics - gfx1100 - 48 GiB	Live workspace
ROCm / HIP	HIP 7.2.53211-e1a6bc5663	PyTorch runtime
PyTorch	2.9.1+rocm7.2.1 - GPU available	Runtime check
Warm SDXL	11.981 s average	13.380, 11.283, 11.279 s
Cold SDXL	17.665 s	Complete bridge round trip
Warm Qwen vision	7.089 s average	7.096, 7.089, 7.081 s
Cold Qwen vision	176.884 s	Includes first model load
Peak VRAM	24.00 GiB	Both services resident

Security verification: the public bridge returned HTTP 401 without credentials and HTTP 200 with the private token. ComfyUI remained loopback-only.

Why this meets the Track 1 criteria

A complete public-safe submission with measured AMD evidence

FUNCTIONAL COMPLETENESS

Body input and a real map scene flow through Radeon inference to a visible, savable result. Image creation and object understanding are both implemented end to end.

PRACTICAL VALUE

Accessible exploration, geographic learning, travel memories, hands-free discovery and creator-ready outputs are grounded in real places rather than abstract prompts.

INNOVATION

Embodied interaction, geospatial media, privacy-preserving local tracking and two complementary Radeon workloads form one coherent product experience.

REPRODUCTION

- Clone the public source repository and copy .env.example to .env.
- Launch the AMD OneClick ROCm workspace with persistent storage.
- Install and start ComfyUI, then start the repository-versioned FastAPI bridge.
- Expose only the authenticated bridge and configure AMD_COMFYUI_* on the app server.
- Run npm run dev, confirm /amd/status, then execute the three demonstrated workflows.
- Run npm run benchmark:amd for reproducible image timing and inspect structured server logs.

SUBMISSION PACKAGE

- Project profile PDF
- Complete public source and README
- Current architecture diagram
- 3-5 minute actual-operation demo plan
- Supplementary poster
- Exact AMD benchmarks and claim boundaries

github.com/EimisPacheco/my-magic-map-agi-adm